

EC200D&ECx00G&EC600U&EG800G Series QuecOpen(SDK) NTP API Reference Manual

LTE Standard Module Series

Version: 1.0

Date: 2023-09-20

Status: Released



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Quectel Wireless Solutions Co., Ltd.

Building 5, Shanghai Business Park Phase III (Area B), No.1016 Tianlin Road, Minhang District, Shanghai 200233, China

Tel: +86 21 5108 6236

Email: info@quectel.com

Or our local offices. For more information, please visit:

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About the Document

Revision History

Version	Date	Author	Description
-	2023-08-03	Kruskal ZHU	Creation of the document
1.0	2023-09-20	Kruskal ZHU	First official release

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1 Introduction

Quectel ECx00G family, EC200D, EC600U and EG800G series modules support QuecOpen® solution. QuecOpen® is an embedded development platform based on RTOS. It is intended to simplify the design and development of IoT applications. For more information on QuecOpen®, see **document [1]** and **document [2]**.

This document is applicable to QuecOpen® solution based on CSDK build environment, and introduces the NTP API, related calling process and the example on Quectel ECx00G family, EC200D, EC600U and EG800G series modules.

1.1. Applicable Modules

Table 1: Applicable Modules

Module Family	Module
-	EC200D Series
-	EC600U Series
ECx00G	EC200G-CN
	EC600G Series
	EC800G-CN
-	EG800G Series

NOTE

For the EC200D series module, the file paths mentioned in this document do not contain *components*.

2 NTP API Calling Process

Before enabling NTP to synchronize time, you need to register the network and start a data call to establish a data channel first. If the data channel has been established in other tasks, you can skip the steps of "Wait for network registration" and "Start a data call". The calling process is shown in the figure below:

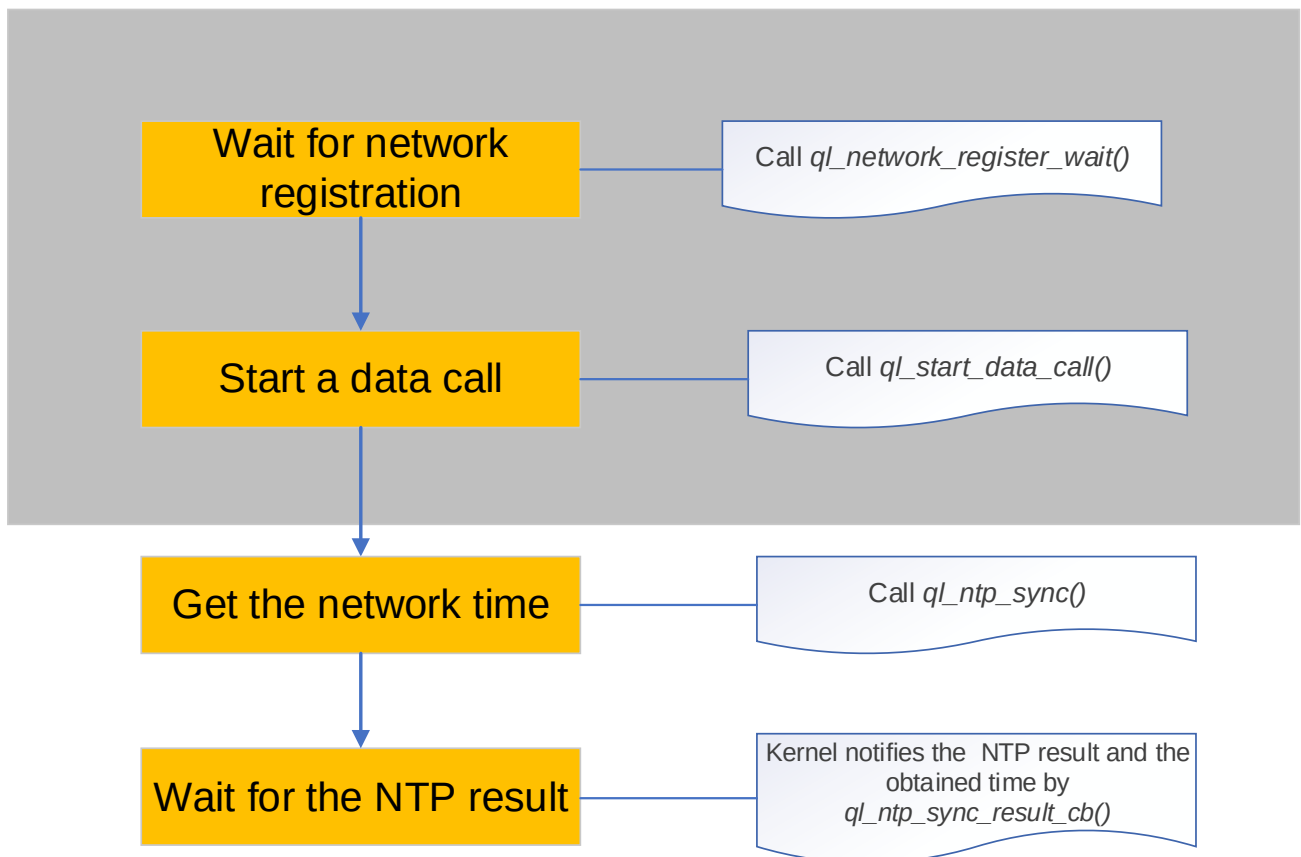


Figure 1: NTP API Calling Process

NOTE

See **document [3]** for details about `ql_network_register_wait()` and `ql_start_data_call()`.

3 NTP API

3.1. Header File

ql_ntp_client.h, the header file of NTP API, is in the `\components\ql-kernel\inc` directory. Unless otherwise specified, all header files mentioned in this document are in this directory.

3.2. Example

The NTP example in application layer is provided in the SDK of the module and NTP client example file is in the `components\ql-application\ntp\ntp_demo.c` directory.

3.3. API Description

3.3.1. ql_ntp_sync

This function enables the feature of NTP synchronization time.

● Prototype

```
ntp_client_id ql_ntp_sync(const char* host, ql_ntp_sync_option *user_option, ql_ntp_sync_result_cb cb, void *arg, int *error_code)
```

● Parameter

host:

[In] The address of NTP server.

user_option:

[In] NTP configuration information. See **Chapter 3.3.1.2** for details.

cb:

[In] Callback function for notifying the NTP synchronization result and the obtained time. See **Chapter 3.3.1.3** for details.

arg:

[In] The callback argument of the NTP synchronization result notification function.

error_code:

[Out] NTP result code called by the function. See **Chapter 3.3.1.4** for details.

● Return Value

NTP client handle. See **Chapter 3.3.1.1** for details.

3.3.1.1. ntp_client_id

NTP client handle is defined below:

```
typedef int ntp_client_id
```

3.3.1.2. ql_ntp_sync_option

The structure of NTP configuration information:

```
typedef struct{
    int    pdp_cid;
    int    sim_id;
    int    retry_cnt;
    int    retry_interval_tm;
}ql_ntp_sync_option
```

● Parameter

Type	Parameter	Description
int	<i>pdp_cid</i>	The PDP channel ID bound to NTP client.
int	<i>sim_id</i>	(U)SIM card bound to NTP client.
int	<i>retry_cnt</i>	The number of retries when NTP fails to get the time.
int	<i>retry_interval_tm</i>	The interval time of NTP request.

3.3.1.3. ql_ntp_sync_result_cb

This function is the event notification function of NTP, notifying the NTP synchronization result and the obtained time.

● Prototype

```
typedef void(*ql_ntp_sync_result_cb)(ntp_client_id cli_id, int result, struct tm *sync_time, void *arg)
```

● Parameter

cli_id:

[In] NTP client handle, obtained by *ql_ntp_sync()*.

result:

[Out] NTP result code. See **Chapter 3.3.1.4** for details.

sync_time:

[Out] Obtained time.

arg:

[In] Callback argument, passed into by *ql_ntp_sync()*.

● Return Value

None

3.3.1.4. ql_ntp_error_code_e

The enumeration of NTP result codes:

```
typedef enum
{
    QL_NTP_SUCCESS = 0,
    QL_NTP_ERROR_UNKNOWN = 550 | (QL_COMPONENT_LWIP_QNTP << 16),
    QL_NTP_ERROR_WODBLOCK = 551 | (QL_COMPONENT_LWIP_QNTP << 16),
    QL_NTP_ERROR_INVALID_PARAM = 552 | (QL_COMPONENT_LWIP_QNTP << 16),
    QL_NTP_ERROR_OUT_OF_MEM = 553 | (QL_COMPONENT_LWIP_QNTP << 16),
    QL_NTP_ERROR_SOCKET_ALLOC_FAIL = 554 | (QL_COMPONENT_LWIP_QNTP << 16),
    QL_NTP_ERROR_SOCKET_SEND_FAIL = 558 | (QL_COMPONENT_LWIP_QNTP << 16),
    QL_NTP_ERROR_SOCKET_RECV_FAIL = 559 | (QL_COMPONENT_LWIP_QNTP << 16),
    QL_NTP_ERROR_OPEN_PDP_FAIL = 561 | (QL_COMPONENT_LWIP_QNTP << 16),
    QL_NTP_ERROR_DNS_FAIL = 565 | (QL_COMPONENT_LWIP_QNTP << 16),
```

```
QL_NTP_ERROR_TIMEOUT = 569 | (QL_COMPONENT_LWIP_QNTP << 16),
}ql_ntp_error_code_e
```

● Member

Member	Description
<i>QL_NTP_SUCCESS</i>	Successful execution.
<i>QL_NTP_ERROR_UNKNOWN</i>	Unknown error(s).
<i>QL_NTP_ERROR_WODBLOCK</i>	NTP request is blocked.
<i>QL_NTP_ERROR_INVALID_PARAM</i>	Invalid parameter(s).
<i>QL_NTP_ERROR_OUT_OF_MEM</i>	Out of memory.
<i>QL_NTP_ERROR_SOCKET_ALLOC_FAIL</i>	Failed to create the socket
<i>QL_NTP_ERROR_SOCKET_SEND_FAIL</i>	Failed to send an NTP request.
<i>QL_NTP_ERROR_SOCKET_RECV_FAIL</i>	Failed to receive an NTP response.
<i>QL_NTP_ERROR_OPEN_PDP_FAIL</i>	Failed to activate PDP.
<i>QL_NTP_ERROR_DNS_FAIL</i>	Failed to parse DNS.
<i>QL_NTP_ERROR_TIMEOUT</i>	NTP request timeout.

4 Appendix References

Table 2: Related Documents

Document Name
[1] Quectel_ECx00G&EC600U&EG800G_Series_QuecOpen(SDK)_CSDK_Quick_Start_Guide
[2] Quectel_EC200D_Series_QuecOpen(SDK)_CSDK_Quick_Start_Guide
[3] Quectel_EC200D&ECx00G&EC600U&EG800G_Series_QuecOpen(SDK)_Data_Call_API_Reference_Manual

Table 3: Terms and Abbreviations

Abbreviation	Description
API	Application Programming Interface
DNS	Domain Name Server
ID	Identifier
IoT	Internet of Things
LTE	Long Term Evolution
NTP	Network Time Protocol
PDP	Packet Data Protocol
SDK	Software Development Kit